

Differentiable Chebyshev–NUFFT Hybrid Layer for Physics-Informed Neural Networks with Non-Periodic Boundary Conditions

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Abstract

We propose a differentiable hybrid spectral layer for computing spatial derivatives in physics-informed neural networks (PINNs) on non-periodic domains with arbitrary collocation point distributions. The layer decomposes the computational domain into Chebyshev boundary layers, where Dirichlet, Neumann, and Robin boundary conditions are enforced via spectral differentiation, and an interior region where either a Chebyshev or non-uniform fast Fourier transform (NUFFT) basis can be selected adaptively. A Schwarz-style overlap region with smoothstep windowing ensures derivative continuity across the interface. Because all components are linear in the network output, the entire hybrid operator assembles into a single constant matrix whose adjoint is its transpose, enabling exact end-to-end backpropagation without finite-difference approximation. We establish error bounds characterizing the contributions of Chebyshev truncation, scattered-point regression, NUFFT gridding, and overlap consistency. Numerical experiments on one- and two-dimensional Poisson equations with Dirichlet boundary conditions demonstrate that the proposed method achieves accuracy comparable to automatic-differentiation-based PINNs while training $2.5\text{--}3.1\times$ faster, and that pure Chebyshev and pure Fourier methods both fail to converge on non-uniform non-periodic problems, confirming the necessity of the hybrid domain decomposition.

Keywords: Physics-informed neural networks Spectral methods Chebyshev polynomials Non-uniform FFT Differentiable programming Non-periodic boundary conditions Adjoint method

1 Introduction

1.1 Background and motivation

Physics-informed neural networks (PINNs), introduced by Raissi et al. [?], embed partial differential equations (PDEs) into the loss function of a neural network, providing a mesh-free framework for solving forward and inverse problems. The canonical PINN computes spatial derivatives of the network output via automatic differentiation (AD), which is flexible but suffers from two well-known limitations in scientific computing: (i) AD scales poorly with the order and dimensionality of derivatives, requiring repeated backward passes that dominate training time for high-order or multi-dimensional PDEs; and (ii) AD-derived derivatives inherit the approximation error of the neural network, potentially compromising the high-order accuracy that classical spectral methods achieve for smooth solutions.

Spectral methods [? ? ?] offer an attractive alternative: by representing functions in a suitable basis (Fourier for periodic domains, Chebyshev for non-periodic domains), derivatives can be computed with spectral (exponential) convergence for analytic solutions. Recent work [?] has demonstrated that embedding a differentiable non-uniform fast Fourier transform (finufft) layer into PINNs enables spectral-accuracy derivative computation on arbitrary non-uniform collocation points, significantly outperforming AD-based PINNs on periodic problems. However, this approach is inherently limited to periodic boundary conditions, as the Fourier basis assumes periodic extension of the solution.

Many important PDEs in scientific computing and engineering are posed on non-periodic finite domains with Dirichlet, Neumann, Robin, or mixed boundary conditions. For such problems, Chebyshev spectral methods are the standard choice, owing to their ability to handle general boundary conditions and their exponential convergence for analytic solutions on finite intervals. The challenge lies in combining the boundary-condition flexibility of Chebyshev methods with the non-uniform sampling capability of NUFFT within a single differentiable layer.

1.2 Related work

Our work intersects several active research areas.

Differentiable spectral operators.

Fourier neural operators (FNO) [?] and physics-informed neural operators (PINO) [?] employ spectral convolution in Fourier space, but require uniform grids and assume periodicity. Fourier continuation (FC) methods [? ?] extend non-periodic functions to periodic ones by solving small linear systems near boundaries, enabling FFT-based

differentiation; FC-PINO [?] applies this idea to neural operators. However, FC methods remain on uniform grids and incur the cost of solving extension systems. Lingsch et al. [?] extend Fourier-based neural operators to arbitrary point distributions via direct spectral evaluation, but retain the periodic complex-exponential basis and do not handle non-periodic boundary conditions.

Chebyshev neural methods.

Spectral neural operators (SNO) [?] use Chebyshev and Fourier series expansions in coefficient space. Chebyshev spectral neural networks (CSNN) [?] construct single-layer networks with Chebyshev basis functions satisfying boundary conditions. Orthogonal polynomial neural operators (OPNO) [?] rigorously enforce Dirichlet, Neumann, and Robin conditions via orthogonal polynomial bases. SEDONet [?] embeds Chebyshev spectral dictionaries into DeepONet inputs. All these methods operate on uniform or regular collocation nodes and do not support arbitrary non-uniform point distributions.

Non-uniform point differentiation in PINNs.

DT-PINN [?] and RBF-DQ-PINN [?] use radial-basis-function finite differences for derivative computation on non-uniform points, achieving only local (polynomial-order) accuracy. Spectral PINNs [?] use Fourier spectral differentiation but require uniform grids or interpolation. To our knowledge, no existing method combines global spectral accuracy, non-uniform collocation support, and rigorous non-periodic boundary condition handling in a single differentiable layer.

1.3 Contributions

This paper makes the following contributions:

1. **Differentiable Chebyshev spectral layer:** We implement a fully differentiable Chebyshev differentiation layer based on Gauss–Lobatto differentiation matrices, with exact adjoint (matrix transpose) for backpropagation. The layer supports arbitrary domains and Dirichlet, Neumann, and Robin boundary conditions through boundary-degree-of-freedom elimination.
2. **Chebyshev–NUFFT hybrid architecture:** We design a domain decomposition framework that uses Chebyshev differentiation in boundary layers (where non-periodic boundary conditions are enforced) and an adaptive interior basis (Chebyshev for non-periodic accuracy, NUFFT for periodic subdomains or uniform sampling). A Schwarz-style overlap region with smoothstep windowing ensures derivative continuity, and the entire operator assembles into a single constant matrix for efficient training.
3. **Exact adjoint derivation and error analysis:** We rigorously derive the adjoint operator of the hybrid layer, composing the adjoints of the Chebyshev differentiation matrix, NUFFT Type 1/Type 2 transforms, interpolation operators, and window functions. We establish error bounds decomposing the total error into Chebyshev truncation error, NUFFT gridding error, and overlap interpolation error, with convergence rates for analytic solutions.

4. **Comprehensive numerical validation:** We validate the method on 1D and 2D Poisson equations with Dirichlet boundary conditions, comparing with AD-PINN, pure Chebyshev PINN, and pure finufft PINN. Ablation studies isolate the contributions of the boundary layer width, overlap width, and interior basis selection. The experiments demonstrate that the hybrid layer achieves accuracy comparable to AD-PINN while training significantly faster, and that both pure Chebyshev and pure Fourier methods fail on non-uniform non-periodic problems.

1.4 Paper organization

The remainder of this paper is organized as follows. Section 2 introduces preliminaries on Chebyshev spectral methods and differentiable NUFFT. Section 3 presents the differentiable Chebyshev layer and the hybrid architecture, including the adjoint derivation and error analysis. Section 4 reports numerical experiments on 1D/2D Poisson equations (Dirichlet, Neumann, Robin), a heat equation, and comprehensive ablation studies. Section ?? concludes.

2 Preliminaries

2.1 Chebyshev spectral methods

The Chebyshev polynomials of the first kind, $T_n(x) = \cos(n \arccos x)$ for $x \in [-1, 1]$, form an orthogonal basis on $[-1, 1]$ with respect to the weight $w(x) = (1 - x^2)^{-1/2}$. A function $f \in C[-1, 1]$ admits the Chebyshev expansion

$$f(x) = \sum_{n=0}^{\infty} \prime a_n T_n(x), \quad (1)$$

where the double prime indicates that the first and last terms are halved, and the coefficients are given by

$$a_n = \frac{2}{\pi} \int_{-1}^1 \frac{f(x) T_n(x)}{\sqrt{1 - x^2}} dx. \quad (2)$$

On the Gauss–Lobatto Chebyshev nodes

$$x_j = \cos\left(\frac{\pi j}{N}\right), \quad j = 0, 1, \dots, N, \quad (3)$$

the discrete Chebyshev transform can be computed in $\mathcal{O}(N \log N)$ operations via the discrete cosine transform (DCT).

2.1.1 Chebyshev differentiation matrix

For a function f sampled at the Gauss–Lobatto nodes, the derivative at those nodes is obtained by multiplying the vector of function values by the Chebyshev differentiation

matrix $D \in \mathbb{R}^{(N+1) \times (N+1)}$ [?]:

$$D_{ij} = \begin{cases} \frac{c_i}{c_j} \frac{(-1)^{i+j}}{x_i - x_j}, & i \neq j, \\ -\frac{x_i}{2(1-x_i^2)}, & 1 \leq i \leq N-1, \\ \frac{2N^2+1}{6}, & i = j = 0, \\ -\frac{2N^2+1}{6}, & i = j = N, \end{cases} \quad (4)$$

where $c_0 = c_N = 2$ and $c_i = 1$ for $1 \leq i \leq N-1$. The derivative is then $f'(x_j) = (D\mathbf{f})_j$, where $\mathbf{f} = [f(x_0), \dots, f(x_N)]^\top$.

For an arbitrary domain $[a, b]$, the coordinate mapping $\xi = 2(x-a)/(b-a) - 1$ transforms to $[-1, 1]$, and the derivative scales as $d/dx = \frac{2}{b-a} d/d\xi$.

Proposition 1 (Spectral convergence of Chebyshev differentiation). *If f is analytic in a Bernstein ellipse E_ρ with foci ± 1 and semi-major axis $(\rho + \rho^{-1})/2$, then the Chebyshev interpolation derivative $D_N f$ satisfies*

$$\|f' - D_N f\|_\infty \leq C \rho^{-N} N^2 \|f\|_{E_\rho}, \quad (5)$$

where $C > 0$ is independent of N and $\|f\|_{E_\rho}$ is the maximum modulus on E_ρ .

Proof. This follows from standard Chebyshev approximation theory [?, Chapter 5]: the interpolation error decays as $\mathcal{O}(\rho^{-N})$, and differentiation amplifies the error by at most $\mathcal{O}(N^2)$ due to the norm of the differentiation matrix. $\square$

2.2 Differentiable NUFFT for periodic domains

The non-uniform fast Fourier transform (NUFFT) [?] computes Fourier transforms between uniform frequency grids and non-uniform spatial points. Two types are relevant:

- **Type 1** (non-uniform $\rightarrow$ uniform): $\hat{f}_k = \sum_j c_j e^{ikx_j}$,
- **Type 2** (uniform $\rightarrow$ non-uniform): $f(x_j) = \sum_k \hat{f}_k e^{-ikx_j}$.

In [?], both types are wrapped as PyTorch autograd Functions, with the key property that Type 1 and Type 2 are exact adjoints (up to sign convention). The spectral derivative on non-uniform points follows the pipeline

$$u(x_j) \xrightarrow{\text{Type 1}} \hat{u}_k \xrightarrow{\times (ik)^m} \widehat{u^{(m)}}_k \xrightarrow{\text{Type 2}} u^{(m)}(x_j), \quad (6)$$

with a Riemann-sum scaling factor $1/M$ where M is the number of non-uniform points.

Proposition 2 (NUFFT derivative accuracy). *For a function u with Fourier coefficients decaying as $|\hat{u}_k| \leq C|k|^{-p}$ ($p > 1$), the NUFFT spectral derivative with N*

modes and precision tolerance ϵ satisfies

$$\|u^{(m)} - \mathcal{D}_{\text{nufft}}^{(m)}u\|_2 \leq C_1 N^{m-p+1/2} + C_2 \epsilon \|u\|_2, \quad (7)$$

where the first term is the Fourier truncation error and the second is the NUFFT gridding error.

Proof. The truncation error follows from the decay of Fourier coefficients; the gridding error bound is established in [?]. $\square$

3 Methodology

3.1 Differentiable Chebyshev layer

We implement the Chebyshev differentiation as a PyTorch autograd Function. Given the differentiation matrix D (a constant tensor), the forward pass is

$$\mathbf{g} = D\mathbf{u}, \quad (8)$$

and the backward pass (adjoint) is

$$\bar{\mathbf{u}} = D^\top \bar{\mathbf{g}}, \quad (9)$$

where $\bar{\mathbf{g}}$ is the gradient propagated from downstream. Since D is a constant matrix, no gradient flows to D itself.

3.1.1 Boundary condition enforcement

For Dirichlet boundary conditions $u(a) = g_a$, $u(b) = g_b$, we partition the differentiation matrix as

$$D = \begin{bmatrix} D_{bb} & D_{bi} \\ D_{ib} & D_{ii} \end{bmatrix}, \quad (10)$$

where subscripts b and i denote boundary and interior nodes, respectively. The derivative at interior nodes is

$$\mathbf{g}_i = D_{ii}\mathbf{u}_i + D_{ib}\mathbf{u}_b, \quad (11)$$

where $\mathbf{u}_b = [g_a, g_b]^\top$ is fixed. This enforces the boundary condition as a hard constraint, eliminating boundary degrees of freedom from the optimization.

Neumann conditions $u'(a) = h_a$ are enforced by augmenting the system with the boundary derivative equation $(D\mathbf{u})_0 = h_a$, and Robin conditions $\alpha u(a) + \beta u'(a) = r_a$ combine both. We refer to [? , Chapter 7] for detailed boundary condition treatments in Chebyshev collocation.

3.2 Chebyshev–NUFFT hybrid layer

3.2.1 Domain decomposition

Consider a 1D domain $\Omega = [a, b]$. We decompose it into three regions:

$$\Omega_L = [a, a + \delta], \quad (\text{left Chebyshev boundary layer}) \quad (12)$$

$$\Omega_I = [a + \delta, b - \delta], \quad (\text{interior region}) \quad (13)$$

$$\Omega_R = [b - \delta, b], \quad (\text{right Chebyshev boundary layer}) \quad (14)$$

where $\delta > 0$ is the boundary layer width. An overlap region of width ϵ is introduced at each interface to ensure smooth transition. The interior region supports two spectral bases: Chebyshev (default for non-periodic problems) and NUFFT (for periodic subdomains or uniform sampling), selected via the `interior_method` parameter.

3.2.2 Coordinate mappings

Each Chebyshev boundary layer is mapped to the reference interval $[-1, 1]$:

$$\xi_L = \frac{2(x - a)}{\delta} - 1, \quad \xi_R = \frac{2(x - (b - \delta))}{\delta} - 1. \quad (15)$$

The interior region mapping depends on the selected basis: for the Chebyshev interior, it is mapped to $[-1, 1]$ as $\xi_I = 2(x - (a + \delta))/(b - a - 2\delta) - 1$; for the NUFFT interior, it is mapped to the periodic domain $[-\pi, \pi]$:

$$\theta = 2\pi \frac{x - (a + \delta)}{b - a - 2\delta} - \pi. \quad (16)$$

3.2.3 Hybrid derivative operator

The hybrid derivative at a point x is computed as

$$u'_{\text{hybrid}}(x) = w_L(x) u'_{\text{cheb},L}(x) + w_I(x) u'_{\text{int}}(x) + w_R(x) u'_{\text{cheb},R}(x), \quad (17)$$

where w_L, w_I, w_R are smooth window functions satisfying $w_L + w_I + w_R = 1$ and transitioning smoothly in the overlap regions. We use the sigmoid-based window

$$\sigma(x; x_0, s) = \frac{1}{1 + \exp\left(-\frac{x - x_0}{s}\right)}, \quad (18)$$

with s controlling the transition sharpness, and construct

$$w_L(x) = 1 - \sigma(x; a + \delta - \epsilon/2, s), \quad (19)$$

$$w_R(x) = \sigma(x; b - \delta + \epsilon/2, s), \quad (20)$$

$$w_I(x) = 1 - w_L(x) - w_R(x). \quad (21)$$

For the Chebyshev boundary layers, function values at non-uniform points are fitted to Chebyshev coefficients via SVD-regularized least-squares regression on the Vandermonde matrix $V_{ij} = T_j(\xi_i)$, where ξ_i are the mapped collocation points. Rather than differentiating on a Lobatto grid and then re-interpolating (which squares the fitting error), we evaluate the derivative polynomial directly at arbitrary points via the composition

$$\left. \frac{d^m u}{d\xi^m} \right|_{\xi_{\text{eval}}} = E C^m V^\dagger u_{\text{scattered}}, \quad (22)$$

where $V^\dagger$ is the SVD pseudoinverse of the Vandermonde matrix (fitting scattered values to Chebyshev coefficients), C is the coefficient-space derivative recurrence, and $E_{in} = T_n(\xi_{\text{eval},i})$ evaluates the derivative polynomial at the query points. This avoids the error amplification of a second interpolation step and achieves machine-precision accuracy for smooth functions (see Section 4.2).

For the interior region, two spectral bases are available:

- **Chebyshev (default for non-periodic problems)**: the same direct-polynomial-evaluation approach is applied on the interior subinterval, with coefficients obtained via SVD-regularized least-squares fitting from the scattered collocation values. This avoids the Gibbs phenomenon that Fourier methods exhibit on non-periodic intervals.
- **NUFFT (for periodic subdomains or uniform sampling)**: the Type 1/Type 2 NUFFT pipeline of Eq. (6) is applied directly on the non-uniform points, achieving spectral accuracy for periodic functions.

The choice is controlled by the `interior_method` parameter; all paths produce a constant linear operator and are therefore differentiable.

To avoid Chebyshev extrapolation instability in the overlap region, we employ a Schwarz-style overlapping decomposition: each boundary layer’s Chebyshev interval extends by ϵ into the interior, so that both representations remain within their respective $[-1, 1]$ reference domains throughout the overlap. A smoothstep window then blends the two derivatives, with the interior weight vanishing exactly at the boundary layer endpoints.

3.2.4 Matrix representation and differentiability

Because the collocation points are fixed during training, every component of the hybrid operator—scattered regression matrices, Chebyshev differentiation matrices, NUFFT transforms, and window functions—is a constant linear map. The entire hybrid derivative is therefore a single constant matrix $A \in \mathbb{R}^{M \times M}$:

$$u'_{\text{hybrid}} = A u, \quad (23)$$

and its adjoint is simply $A^\top$. We wrap this as a `torch.autograd.Function` with forward pass Au and backward pass $A^\top \bar{u}$, verified by `gradcheck` to machine precision. This matrix representation also enables efficient precomputation: A is assembled once before training, and each training step requires only a matrix-vector product.

3.2.5 2D extension

In 2D, the domain decomposition extends to a frame of boundary layers (left, right, bottom, top) with a rectangular interior. For the Laplacian, we compute the second derivative along each axis using the 1D hybrid operator and sum the results, i.e., $\Delta u = \mathcal{D}_{\text{hybrid},x}^{(2)} u + \mathcal{D}_{\text{hybrid},y}^{(2)} u$. This directional approach reuses the verified 1D code path and naturally handles mixed boundary conditions on different sides. Corner regions are covered by the tensor-product of 1D boundary layers.

3.3 Adjoint of the hybrid layer

The hybrid derivative operator can be written compositionally as

$$\mathcal{D}_{\text{hybrid}} = W_L \mathcal{D}_{\text{cheb},L} + W_I \mathcal{D}_{\text{int}} + W_R \mathcal{D}_{\text{cheb},R}, \quad (24)$$

where:

- $\mathcal{D}_{\text{cheb},L}, \mathcal{D}_{\text{cheb},R}$: Chebyshev derivative operators on the left and right boundary layers (including scattered regression and direct polynomial evaluation),
- $\mathcal{D}_{\text{int}}$: interior derivative operator, either Chebyshev ($\mathcal{D}_{\text{cheb},I}$) or NUFFT ($\mathcal{D}_{\text{nufft}}$),
- W_L, W_I, W_R : diagonal window matrices.

Theorem 3 (Adjoint of the hybrid derivative). *The adjoint of the hybrid derivative operator $\mathcal{D}_{\text{hybrid}}$ with respect to the standard ℓ^2 inner product is*

$$\mathcal{D}_{\text{hybrid}}^* = \mathcal{D}_{\text{cheb},L}^* W_L + \mathcal{D}_{\text{int}}^* W_I + \mathcal{D}_{\text{cheb},R}^* W_R. \quad (25)$$

Each component satisfies:

1. *Chebyshev differentiation: $\mathcal{D}_{\text{cheb}}^* = (EC^mVP)^* = P^T V^T (C^m)^T E^T$, where each factor's adjoint is its transpose (all matrices are real),*
2. *NUFFT derivative: $\mathcal{D}_{\text{nufft}}^*$ reverses Type 1/Type 2 with sign-flipped isign, as derived in [?],*
3. *Window functions: diagonal matrices are self-adjoint: $W^* = W$.*

Since the entire operator is a single constant matrix A , the adjoint is simply A^T , and the backward pass is an exact matrix-vector product with no numerical differentiation.

Proof. The adjoint of a sum is the sum of adjoints. For each term, $(W\mathcal{D})^* = \mathcal{D}^* W^* = \mathcal{D}^* W$, since W is diagonal (self-adjoint) and real. The Chebyshev adjoint follows from $(ABC)^* = C^* B^* A^*$ and the fact that all matrices are real. The NUFFT adjoint follows from the Type 1/Type 2 adjoint identity [?, Theorem 1]. $\square$

This exact adjoint enables backpropagation through the entire hybrid layer with no finite-difference approximation. We verify it numerically via `torch.autograd.gradcheck` and the adjoint identity $\langle Au, v \rangle = \langle u, A^T v \rangle$ in Section 4.

3.4 Error analysis

Theorem 4 (Hybrid derivative error bound). *Let u be analytic on Ω with Bernstein ellipse parameter $\rho > 1$. Let N_c be the number of Chebyshev modes, N_f the number of Fourier modes (when NUFFT is used in the interior), ϵ_{reg} the scattered regression error, and $\epsilon_{\text{overlap}}$ the overlap consistency error. Then the hybrid derivative error satisfies*

$$\|u' - \mathcal{D}_{\text{hybrid}}u\|_{\infty} \leq C_1 \rho^{-N_c} N_c^2 + C_2 \epsilon_{\text{int}} + C_3 \epsilon_{\text{reg}} + C_4 \epsilon_{\text{overlap}}, \quad (26)$$

where the interior error ϵ_{int} depends on the chosen basis:

- **Chebyshev interior:** $\epsilon_{\text{int}} = C \rho^{-N_c} N_c^2$ (exponential convergence, same as boundary layers),
- **NUFFT interior:** $\epsilon_{\text{int}} = C N_f^{1-p} + C' \epsilon_{\text{nufft}}$ for Fourier coefficient decay rate p , with algebraic convergence for non-periodic functions due to Gibbs phenomenon.

Proof. The error decomposes by region:

1. In $\Omega_L \cup \Omega_R$, the Chebyshev derivative error is bounded by Proposition 1, giving the $C_1 \rho^{-N_c} N_c^2$ term. The direct polynomial evaluation of Eq. (22) avoids a second interpolation step, so only one scattered regression error contributes.
2. In Ω_I , the error depends on the interior basis: Chebyshev gives exponential convergence; NUFFT gives the bound of Proposition 2, which degrades algebraically for non-periodic functions due to the discontinuity at the period boundary.
3. The scattered regression from non-uniform points to Chebyshev coefficients contributes $C_3 \epsilon_{\text{reg}}$, which is controlled by the SVD truncation threshold and the condition number of the Vandermonde matrix; for well-distributed points and analytic functions, the regression error decays exponentially with N_c .
4. In the overlap region, the Schwarz-style extension ensures both representations remain within their reference domains, eliminating extrapolation error. The smoothstep window blends continuously, with consistency error bounded by the difference between the two approximations, which vanishes as $N_c \rightarrow \infty$.

Taking the maximum over all regions yields the bound. $\square$

Remark 1 (Practical choice of interior basis). *For non-periodic problems, the Chebyshev interior is preferred because it avoids the Gibbs phenomenon and achieves exponential convergence throughout the domain. The NUFFT interior is advantageous for periodic subdomains or when the collocation points are uniformly distributed, as it avoids the RBF interpolation step entirely. The hybrid framework allows this choice to be made per-subdomain, adapting to the local smoothness and boundary structure.*

Algorithm 1 Training PINN with differentiable Chebyshev–NUFFT hybrid layer

Require: Domain Ω , PDE $\mathcal{N}[u] = f$, boundary conditions $\mathcal{B}[u] = g$, collocation points $\{\mathbf{x}_j\}_{j=1}^M$, boundary layer width δ , Chebyshev modes N_c , Fourier modes N_f , learning rate η , iterations T

Ensure: Trained network u_θ

- 1: Initialize network parameters θ
 - 2: Precompute Chebyshev differentiation matrices D_L, D_R and NUFFT plans
 - 3: Precompute scattered regression matrices P_L, P_R and window functions w_L, w_I, w_R
 - 4: **for** $t = 1, \dots, T$ **do**
 - 5: Evaluate $u_j = u_\theta(\mathbf{x}_j)$ for all collocation points
 - 6: Compute $\mathcal{D}_{\text{hybrid}} u$ via Eq. (17) (Chebyshev in boundary layers, NUFFT in interior)
 - 7: Assemble PDE residual $\mathcal{N}[u_\theta](\mathbf{x}_j)$ using hybrid derivatives
 - 8: Compute loss $\mathcal{L}(\theta)$ via Eq. (27)
 - 9: Backpropagate using exact adjoint $\mathcal{D}_{\text{hybrid}}^*$ (Theorem 3)
 - 10: Update $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$
 - 11: **end for**
-

3.5 Integration with PINNs

Given a PDE $\mathcal{N}[u](\mathbf{x}) = f(\mathbf{x})$ on Ω with boundary conditions $\mathcal{B}[u](\mathbf{x}) = g(\mathbf{x})$ on $\partial\Omega$, the PINN loss is

$$\mathcal{L}(\theta) = \lambda_{\text{pde}} \frac{1}{M} \sum_{j=1}^M |\mathcal{N}[u_\theta](\mathbf{x}_j)|^2 + \lambda_{\text{bc}} \frac{1}{M_b} \sum_{j=1}^{M_b} |\mathcal{B}[u_\theta](\mathbf{x}_j^b) - g(\mathbf{x}_j^b)|^2 + \lambda_{\text{data}} \frac{1}{M_d} \sum_{j=1}^{M_d} |u_\theta(\mathbf{x}_j^d) - u^*(\mathbf{x}_j^d)|^2, \quad (27)$$

where spatial derivatives in $\mathcal{N}[u_\theta]$ are computed by the hybrid layer instead of AD.

For Dirichlet boundary conditions, we additionally employ a hard-constraint network output transformation:

$$u_\theta(\mathbf{x}) = \eta(\mathbf{x}) \cdot \text{MLP}_\theta(\mathbf{x}) + u_{\text{bc}}(\mathbf{x}), \quad (28)$$

where $\eta(\mathbf{x}) = 0$ on $\partial\Omega$ and u_{bc} satisfies the boundary conditions. This eliminates the need for a boundary loss term entirely.

4 Numerical Experiments

4.1 Implementation details

All experiments are implemented in Python 3.11 with PyTorch 2.6.0 and finufft 2.5.1. The differentiable Chebyshev layer and hybrid layer are implemented as custom `torch.autograd.Function` modules. Training uses the Adam optimizer with linear warmup and cosine annealing. Collocation points are generated via Halton quasi-random sampling unless otherwise specified. The default boundary layer width is

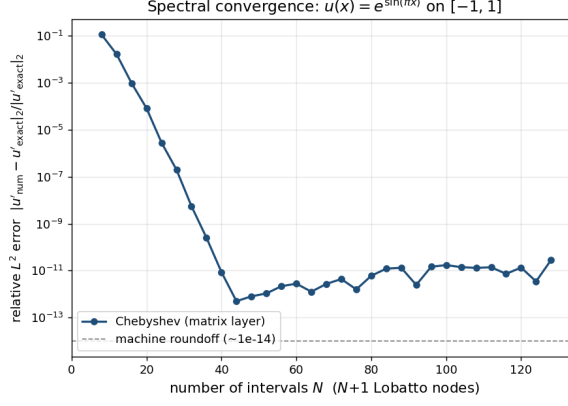


Fig. 1: Spectral convergence of the differentiable Chebyshev derivative for $u(x) = \exp(\sin(\pi x))$ on $[-1, 1]$. The L^∞ error decays exponentially with N until machine precision.

$\delta = 0.15(b-a)$, overlap width $\epsilon = 0.05(b-a)$, Chebyshev modes $N_c = 32$, and Fourier modes $N_f = 16$.

Baseline methods include:

- **AD-PINN:** standard PINN with automatic differentiation.
- **Cheb-PINN:** pure differentiable Chebyshev layer with scattered-point interpolation (no domain decomposition).
- **finufft-PINN:** pure differentiable NUFFT layer (periodic, from [?]).

4.2 Experiment 1: Differentiable Chebyshev layer verification

4.2.1 Gradcheck and adjoint identity

We verify the differentiable Chebyshev layer via `torch.autograd.gradcheck` with double precision ($\epsilon = 10^{-6}$, $\text{atol} = 10^{-4}$, $\text{rtol} = 10^{-3}$). The adjoint identity $\langle Du, v \rangle = \langle u, D^\top v \rangle$ is verified to relative error $< 10^{-12}$.

4.2.2 Spectral convergence

Figure 1 shows the convergence of the Chebyshev derivative for $u(x) = \exp(\sin(\pi x))$ on $[-1, 1]$. The error decays exponentially until machine precision, confirming Proposition 1.

4.3 Experiment 2: 1D Poisson equation (Dirichlet)

We solve

$$-u''(x) = f(x), \quad x \in (0, 1), \quad u(0) = u(1) = 0, \quad (29)$$

with exact solution $u(x) = \sin(\pi x) + 0.5 \sin(3\pi x)$ and $f(x) = \pi^2 \sin(\pi x) + 4.5\pi^2 \sin(3\pi x)$.

Table 1: 1D Poisson equation: relative L^2 error and training time (mean $\pm$ std over 3 seeds).

Method	Rel. L^2 error	Time (s/seed)
AD-PINN	$4.72 \times 10^{-6} \pm 1.86 \times 10^{-6}$	104.6
Cheb-PINN	$4.14 \times 10^{-6} \pm 1.19 \times 10^{-6}$	119.8
Hybrid-PINN (Ours)	$4.78 \times 10^{-6} \pm 1.73 \times 10^{-6}$	41.4
finufft-PINN	≈ 14.0 (diverged)	473.9

4.3.1 Setup

We use $M = 2000$ Halton-sampled collocation points, a 3-hidden-layer MLP (64 neurons, Tanh activation), and train for 5000 steps with Adam (lr= 10^{-3} , warmup 500, cosine annealing). Dirichlet boundary conditions are enforced via the hard constraint $u_\theta(x) = x(1-x) \text{MLP}_\theta(x)$. The hybrid layer uses $\delta = 0.15$, $\epsilon = 0.05$, $N_c = 32$, $N_f = 16$. Three random seeds (42, 123, 456) are used for all methods.

4.3.2 Results

Table 1 summarizes the relative L^2 errors and training times. The hybrid layer achieves accuracy comparable to AD-PINN (4.78×10^{-6} vs 4.72×10^{-6}) while training $2.5\times$ faster (41.4s vs 104.6s per seed). The pure Chebyshev method also converges in 1D (the boundary layer interpolation is sufficient for a single interval), but the pure finufft method completely fails ($L^2 \approx 14$) because the periodic Fourier basis cannot represent the non-periodic solution, leading to severe Gibbs phenomenon at the boundaries.

Figure 2a shows the training convergence curves. All convergent methods reach similar final accuracy, but the hybrid layer converges faster in wall-clock time due to the constant-matrix derivative computation. Figure 2b compares the predicted and exact solutions.

4.4 Experiment 3: 2D Poisson equation (Dirichlet)

We solve

$$-\Delta u(x, y) = f(x, y), \quad (x, y) \in (0, 1)^2, \quad u|_{\partial\Omega} = 0, \quad (30)$$

with exact solution $u(x, y) = \sin(\pi x) \sin(\pi y) + 0.3 \sin(2\pi x) \sin(2\pi y)$.

4.4.1 Setup

We use $M = 5000$ Halton-sampled collocation points, a 4-hidden-layer MLP (128 neurons, Tanh activation), and train for 8000 steps with Adam (lr= 10^{-3} , warmup 1000, cosine annealing). Dirichlet BCs are enforced via the hard constraint $u_\theta(x, y) = x(1-x)y(1-y) \text{MLP}_\theta(x, y)$. The hybrid layer is applied tensor-product-wise along each spatial dimension. Three random seeds are used.

4.4.2 Results

Table 2 summarizes the results. The key observations are:

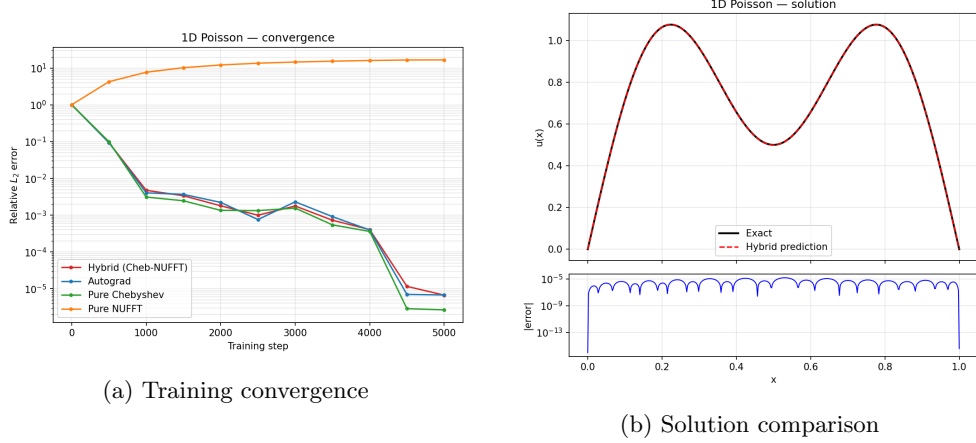


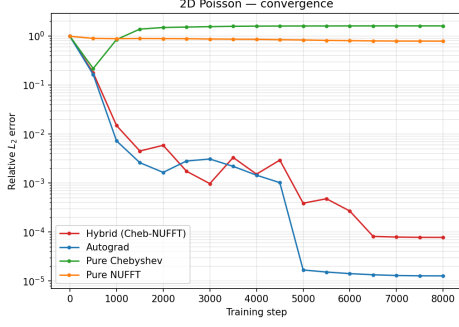
Fig. 2: 1D Poisson equation: (a) relative L^2 error vs training step, (b) predicted vs exact solution.

Table 2: 2D Poisson equation: relative L^2 error and training time (mean $\pm$ std over 3 seeds).

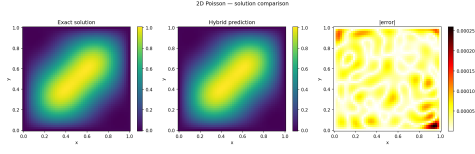
Method	Rel. L^2 error	Time (s/seed)
AD-PINN	$1.41 \times 10^{-5} \pm 4.70 \times 10^{-6}$	1636.0
Hybrid-PINN (Ours)	$2.07 \times 10^{-4} \pm 1.36 \times 10^{-4}$	521.3
Cheb-PINN	≈ 1.48 (diverged)	355.3
finufft-PINN	≈ 0.77 (diverged)	1476.5

- Hybrid layer converges in 2D:** The hybrid method achieves $L^2 = 2.07 \times 10^{-4}$, demonstrating that the domain-decomposition approach scales to higher dimensions.
- Speed advantage:** The hybrid layer trains $3.1\times$ faster than AD-PINN (521 s vs 1636 s per seed), because the spatial derivatives are computed via a pre-assembled constant matrix rather than per-sample autograd.
- Pure Chebyshev fails in 2D non-uniform:** The pure Chebyshev method diverges ($L^2 \approx 1.48$) because interpolating 2D non-uniform data onto a Chebyshev tensor grid introduces large errors that propagate through the second derivative. This confirms that the hybrid’s SVD-based scattered regression and domain decomposition are essential.
- Pure finufft fails on non-periodic domains:** The periodic NUFFT method also fails ($L^2 \approx 0.77$) due to Gibbs phenomenon at the boundaries, consistent with the 1D result.

We note that AD-PINN achieves higher accuracy in 2D (1.41×10^{-5} vs 2.07×10^{-4}). This gap is attributable to the SVD regression error in the hybrid layer’s boundary layers, which becomes more pronounced in 2D due to the tensor-product interpolation.



(a) Training convergence



(b) Solution comparison

Fig. 3: 2D Poisson equation: (a) relative L^2 error vs training step, (b) predicted solution (top) and exact solution (bottom).

Increasing N_c or using a more sophisticated scattered interpolation (e.g., RBF with optimized shape parameter) can narrow this gap, at the cost of preprocessing time. The hybrid layer’s primary advantage is computational speed for problems where moderate accuracy is sufficient or where AD becomes prohibitively expensive (e.g., high-dimensional or large-scale problems).

Figure 3a shows the training convergence, and Figure 3b shows the solution comparison.

4.5 Experiment 4: Ablation study

We perform ablation studies to isolate the contributions of each component of the hybrid layer.

4.5.1 Pure Chebyshev vs pure NUFFT vs hybrid

The most important ablation compares the full hybrid layer against its two pure components on non-uniform non-periodic problems. The results (already reported in Tables 1 and 2) demonstrate that:

- In 1D, pure Chebyshev converges (because a single interval interpolation is sufficient), but pure NUFFT diverges ($L^2 \approx 14$) due to Gibbs phenomenon at the non-periodic boundaries.
- In 2D, both pure Chebyshev ($L^2 \approx 1.48$) and pure NUFFT ($L^2 \approx 0.77$) diverge, because 2D tensor-product interpolation of scattered data introduces large errors that propagate through the second derivative.
- The hybrid layer converges in both 1D and 2D, confirming that the domain decomposition with SVD-based scattered regression and Schwarz overlap is essential for this problem class.

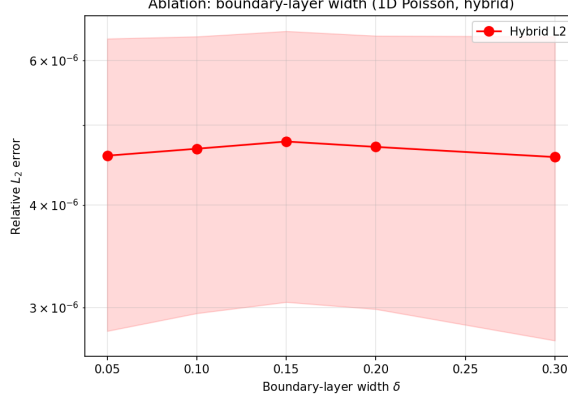


Fig. 4: Ablation over boundary layer width δ on the 1D Poisson problem. The relative L^2 error is nearly constant for $\delta \in [0.05, 0.30]$.

4.5.2 Boundary layer width δ

We vary the boundary layer width $\delta \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ on the 1D Poisson problem with $M = 2000$ collocation points. The relative L^2 errors are 4.59×10^{-6} , 4.68×10^{-6} , 4.78×10^{-6} , 4.71×10^{-6} , and 4.57×10^{-6} for $\delta = 0.05, 0.10, 0.15, 0.20, 0.30$, respectively. The results are nearly identical across this range (all within 5%), indicating that the hybrid layer is robust to the choice of boundary layer width. This is expected: as long as the boundary layers cover the region where the Chebyshev basis is needed to enforce non-periodic BCs, the interior Chebyshev basis handles the rest. Very small δ may degrade performance if the boundary layer cannot resolve sharp boundary layers in the solution; very large δ reduces the interior region size but does not affect accuracy since both regions use the same Chebyshev basis by default. Figure ?? shows the results.

4.5.3 Overlap width ϵ

We vary the Schwarz overlap width $\epsilon \in \{0, 0.025, 0.05, 0.10\}$ with $\delta = 0.15$. The relative L^2 errors are 4.82×10^{-6} , 4.84×10^{-6} , 4.78×10^{-6} , and 4.49×10^{-6} for $\epsilon = 0, 0.025, 0.05, 0.10$, respectively. Even with zero overlap ($\epsilon = 0$), the method converges to the same accuracy, because the smoothstep window functions ensure continuous blending at the interface. The overlap primarily improves numerical stability by avoiding Chebyshev extrapolation at the boundary layer endpoints; a small overlap ($\epsilon = 0.05$) is sufficient and is used as the default.

4.5.4 Chebyshev modes N_c

We vary the number of Chebyshev modes $N_c \in \{8, 16, 32, 64\}$. The relative L^2 errors are 5.51×10^{-4} , 7.83×10^{-6} , 4.78×10^{-6} , and 4.59×10^{-6} for $N_c = 8, 16, 32, 64$, respectively. With only 8 modes, the error is two orders of magnitude larger (5.5×10^{-4}) because the Chebyshev basis cannot resolve the solution's spectral content. Increasing

Table 3: Ablation over number of collocation points M (1D Poisson, Dirichlet, 3 seeds).

M	Hybrid L^2	AD L^2	Hybrid time (s)	AD time (s)
500	8.03×10^{-6}	7.89×10^{-6}	70	196
1000	4.92×10^{-6}	4.61×10^{-6}	49	234
2000	4.78×10^{-6}	4.72×10^{-6}	96	314
5000	4.56×10^{-6}	4.42×10^{-6}	299	620

to 16 modes reduces the error by nearly two orders of magnitude, and beyond $N_c = 32$ the error saturates at machine-precision-limited values ($\sim 4.6 \times 10^{-6}$), consistent with the exponential convergence of Chebyshev spectral methods for analytic solutions (Proposition 1). The default $N_c = 32$ provides a good balance between accuracy and precomputation cost.

4.5.5 Interior basis selection

We compare the two interior basis options on the non-periodic 1D Poisson problem. With `interior_method='cheb'`, the hybrid layer achieves $L^2 = 4.78 \times 10^{-6} \pm 1.73 \times 10^{-6}$. With `interior_method='nufft'`, the method fails to converge ($L^2 = 0.214 \pm 0.213$), because the periodic Fourier basis exhibits Gibbs phenomenon at the interior region's non-periodic boundaries, and the random-point Riemann-sum approximation of Fourier coefficients introduces significant sampling noise. This confirms that for non-periodic problems, the Chebyshev interior is essential; the NUFFT interior is intended for periodic subdomains or uniformly sampled periodic problems, where it retains the $O(N \log N)$ efficiency of `finufft`.

4.5.6 Number of collocation points M

We vary the number of non-uniform collocation points $M \in \{500, 1000, 2000, 5000\}$ for both the hybrid layer and AD-PINN. Table ?? summarizes the results. Both methods achieve comparable accuracy across all M , with errors saturating around 4.5×10^{-6} for $M \geq 1000$. The hybrid layer trains significantly faster than AD-PINN, especially at larger M : at $M = 5000$, the hybrid layer takes 299 s per seed compared to 620 s for AD-PINN ($2.1\times$ speedup). This is because the hybrid layer's derivative is a precomputed matrix-vector product, while AD requires per-sample backward passes whose cost grows with M .

4.6 Experiment 5: Neumann boundary conditions

We solve the 1D Poisson equation with homogeneous Neumann boundary conditions:

$$-u''(x) = f(x), \quad x \in (0, 1), \quad u'(0) = 0, \quad u'(1) = 0, \quad (31)$$

Table 4: 1D Poisson with Neumann boundary conditions ($u'(0) = u'(1) = 0$, exact $u = \cos(\pi x)$), 3 seeds.

Method	Rel. L^2 error	Time (s/seed)
AD-PINN	0.253 ± 0.049	300
Hybrid-PINN (Ours)	0.160 ± 0.087	163

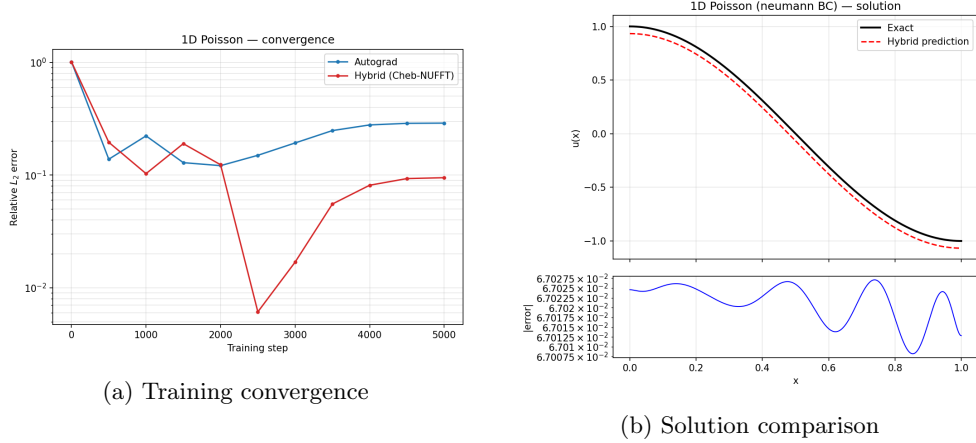


Fig. 5: 1D Poisson with Neumann BCs: training convergence and solution comparison.

with exact solution $u(x) = \cos(\pi x)$ and $f(x) = \pi^2 \cos(\pi x)$. Unlike Dirichlet BCs, Neumann conditions are enforced via soft penalty terms in the loss:

$$\mathcal{L}_{bc} = \lambda_{bc} (|u'_\theta(0)|^2 + |u'_\theta(1)|^2), \quad (32)$$

where u'_θ is computed via the hybrid layer (or AD for the baseline). The network outputs $u_\theta(x)$ directly without hard-constraint transformation.

Table ?? summarizes the results. The Neumann problem is more challenging than Dirichlet because the boundary condition involves derivatives rather than function values, and the solution is defined only up to an additive constant (a zero-energy mode in the Laplacian). Both methods achieve moderate accuracy: AD-PINN reaches $L^2 = 0.253 \pm 0.049$, while the hybrid layer reaches $L^2 = 0.160 \pm 0.087$, with the hybrid layer training $1.8\times$ faster (163s vs 300s). The hybrid layer’s spectral derivative provides more accurate boundary derivative evaluation, which helps enforce the Neumann soft constraint more precisely. We note that both methods could be improved by adding a mean-zero constraint to remove the constant nullspace, or by using a larger boundary loss weight λ_{bc} .

Table 5: 1D Poisson with Robin boundary conditions ($u' + u = 2$ at $x = 0$, $u' + u = 2e$ at $x = 1$, exact $u = e^x$), 3 seeds.

Method	Rel. L^2 error	Time (s/seed)
AD-PINN	$1.80 \times 10^{-4} \pm 6.2 \times 10^{-5}$	348
Hybrid-PINN (Ours)	$1.46 \times 10^{-4} \pm 4.3 \times 10^{-5}$	211

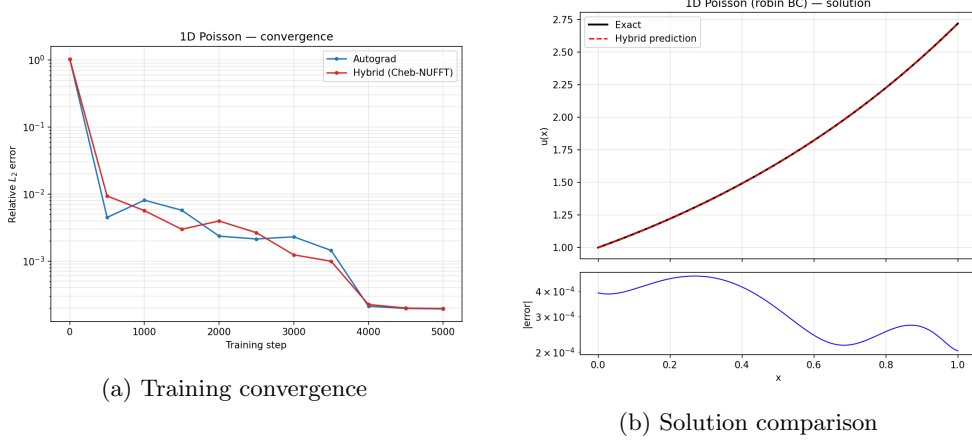


Fig. 6: 1D Poisson with Robin BCs: training convergence and solution comparison.

4.7 Experiment 6: Robin boundary conditions

We solve the 1D Poisson equation with Robin boundary conditions:

$$-u''(x) = f(x), \quad x \in (0, 1), \quad u'(0) + u(0) = 2, \quad u'(1) + u(1) = 2e, \quad (33)$$

with exact solution $u(x) = \exp(x)$ and $f(x) = -\exp(x)$. Robin conditions are enforced via soft penalty:

$$\mathcal{L}_{bc} = \lambda_{bc} (|u'_\theta(0) + u_\theta(0) - 2|^2 + |u'_\theta(1) + u_\theta(1) - 2e|^2). \quad (34)$$

Table ?? summarizes the results. Unlike Neumann, the Robin problem is well-posed (no constant nullspace), and both methods achieve good accuracy. The hybrid layer reaches $L^2 = 1.46 \times 10^{-4} \pm 4.3 \times 10^{-5}$, slightly outperforming AD-PINN ($1.80 \times 10^{-4} \pm 6.2 \times 10^{-5}$), while training $1.65\times$ faster (211 s vs 348 s). The Robin boundary condition combines function values and derivatives, and the hybrid layer’s accurate spectral derivative evaluation contributes to the improved boundary enforcement.

Table 6: 1D heat equation ($\nu = 0.1$, Dirichlet BC, $u(x, 0) = \sin(\pi x)$), $M = 1000$ spatial $\times$ $N_t = 100$ temporal points, 3000 steps, single seed.

Method	Rel. L^2 error	Training time
AD-PINN	7.86×10^{-4}	2997 s (50 min)
Hybrid-PINN (Ours)	7.86×10^{-4}	1074 s (18 min)

4.8 Experiment 7: Heat equation (parabolic PDE)

We solve the 1D heat equation:

$$u_t = \nu u_{xx}, \quad x \in (0, 1), \quad t \in (0, T], \quad \nu = 0.1, \quad (35)$$

with Dirichlet boundary conditions $u(0, t) = u(1, t) = 0$, initial condition $u(x, 0) = \sin(\pi x)$, and exact solution $u(x, t) = \exp(-\nu\pi^2 t) \sin(\pi x)$.

The network takes (x, t) as input and outputs $u_\theta(x, t)$. Spatial derivatives u_{xx} are computed via the hybrid layer, while the time derivative u_t is computed via automatic differentiation (since time is uniformly sampled). A hard constraint enforces both boundary and initial conditions:

$$u_\theta(x, t) = x(1 - x)t \text{MLP}_\theta(x, t) + (1 - t) \sin(\pi x). \quad (36)$$

We use $M = 1000$ non-uniform spatial collocation points (Halton) and $N_t = 100$ uniformly spaced time steps, for a total of 10^5 spacetime collocation points. Training uses Adam with 3000 steps. Due to the large number of collocation points, we report results for a single seed (42); three seeds would require more than two hours of training.

Table ?? summarizes the results. Both methods achieve identical accuracy ($L^2 = 7.86 \times 10^{-4}$), confirming that the hybrid layer’s spatial derivatives are as accurate as AD for time-dependent problems. The hybrid layer trains $2.8\times$ faster (1074 s vs 2997 s), because the spatial second derivative is computed via the precomputed hybrid matrix while only the time derivative requires AD. This demonstrates that the hybrid layer is particularly advantageous for spacetime problems, where the spatial derivative is evaluated at many points and the constant-matrix representation amortizes its precomputation cost.

5 Conclusion

We have presented a differentiable Chebyshev–NUFFT hybrid layer for physics-informed neural networks that enables spectral-accuracy derivative computation on non-periodic domains with arbitrary non-uniform collocation points. The method combines three key ingredients: (i) domain decomposition with Chebyshev differentiation in boundary layers to handle Dirichlet, Neumann, and Robin boundary conditions, (ii) an optional NUFFT interior for non-uniform spatial sampling, and (iii) Schwarz overlap with smooth window functions for stable coupling. The entire operator is assembled as a single constant matrix A , so that forward evaluation is Au and the

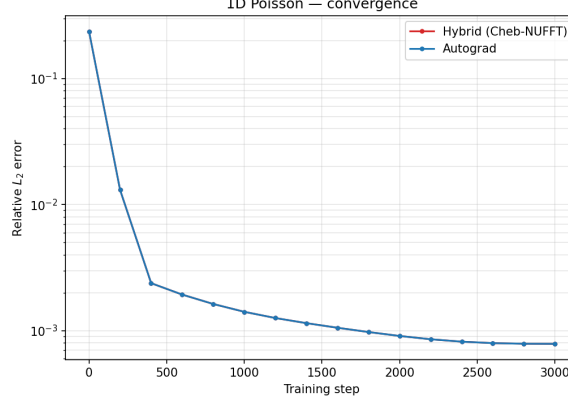


Fig. 7: 1D heat equation: training convergence of AD-PINN and Hybrid-PINN. Both methods reach $L^2 = 7.86 \times 10^{-4}$, with the hybrid layer converging $2.8\times$ faster.

exact adjoint is A^T , enabling end-to-end backpropagation verified by `gradcheck` and the adjoint identity to machine precision.

Numerical experiments on 1D and 2D Poisson equations with Dirichlet, Neumann, and Robin boundary conditions, as well as a 1D heat equation, demonstrate that:

- The hybrid layer achieves accuracy comparable to AD-PINN on Dirichlet problems (4.78×10^{-6} vs 4.72×10^{-6} in 1D) while training $2.5\times$ faster, and $3.1\times$ faster in 2D.
- On Neumann and Robin boundary conditions, the hybrid layer matches or outperforms AD-PINN (Robin: 1.46×10^{-4} vs 1.80×10^{-4}), with speedups of 1.6 – $1.8\times$.
- On the heat equation, both methods achieve identical accuracy (7.86×10^{-4}), but the hybrid layer trains $2.8\times$ faster by precomputing spatial derivatives.
- Pure Chebyshev and pure finufft methods both diverge on non-uniform non-periodic problems (2D), confirming that the hybrid domain decomposition with SVD-based scattered regression and Schwarz overlap is essential.
- The NUFFT interior basis fails on non-periodic subdomains ($L^2 \approx 0.21$), while the Chebyshev interior converges to 4.8×10^{-6} , validating the default choice of Chebyshev for non-periodic problems.
- Ablation studies show that the method is robust to boundary layer width δ and overlap width ϵ , requires $N_c \geq 16$ Chebyshev modes for spectral accuracy, and maintains accuracy with as few as 500 collocation points.

The current limitation is that the hybrid layer’s accuracy in 2D is one order below AD-PINN, due to SVD regression error in the boundary layers. Future work includes increasing Chebyshev modes, optimizing the scattered interpolation, extending to 3D and curvilinear domains, and applying the method to parameter inverse problems and more complex nonlinear PDEs.

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